

ASHLEY (YE) GAO

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📍 540 Landrum Dr Room 2381, Williamsburg, VA 23185

EMPLOYMENT

William & Mary

2023 - Present

Assistant Professor in Computer Science

EDUCATION

University of Virginia

2019 - 2023

Advisor: John A. Stankovic

Thesis: *Addressing Realisms Faced by Deep Learning Models in Cyber Physical Systems*

Ph.D. in Computer Science

University of Virginia

2017 - 2019

M.S. in Computer Science

University of California, San Diego

2012 - 2017

B.S. in Computer Science

B.A. in Literatures of the World

RESEARCH STATEMENT

My research lies at the intersection of **affective computing**, **audio-language modeling**, and **robust machine learning**. I develop methods for **emotion-aware representation learning and generation**, with a focus on audio large language models, controllable text-to-speech, and speech emotion recognition. My work combines advances in **representation learning**, **generative modeling** with applications in **computational healthcare** and **human-centered AI**.

SERVICES

Editorial Roles

S5. Associate Editor. Elsevier Smart Health.

2024 - Present

Grant Review Panels (Invited)

S4. Panelist. National Institutes of Health (NIH).

2025

S3. Panelist. National Institutes of Health (NIH).

2024

S2. Panelist. National Science Foundation (NSF).

2024

Conference Leadership

S1. Demo & Poster Chair. IEEE/ACM CHASE (Connected Health: Applications, Systems and Engineering Technologies).

2024

SELECTED ACCEPTED PUBLICATIONS

Conference and Journal Publications

[ICML'26] Du, H., Shi, J., Lu, S., Zhou, G., **Gao, Y.**, "Sparse Autoencoders for Interpretable Emotion Control in Text-to-Speech." Proceedings of the International Conference on Machine Learning (ICML), 2026.

[ACL'26] Nair, T., Nafee, M. W., Jiang, M., **Gao, Y.**, Chen, H., Zhang, Y., "Confidence-Aware Ranker Ensembles for Robust In-Context Knowledge Editing." Findings of the Association for Computational Linguistics (ACL), 2026.

- [**ACL'26**] Shi, J., Du, H., Song, X., Hong, Y. A., Zhang, Y., Gao, Y., "AffectCodec: Emotion-Preserving Neural Speech Codec for Expressive Speech Modeling." Findings of the Association for Computational Linguistics (ACL), 2026.
- [**ICASSP'26**] Shi, J., Du, H., He, Y., Hong, Y. A., Gao, Y., "Emotion-Aligned Generation in Diffusion Text-to-Speech Models via Preference-Guided Optimization." IEEE ICASSP, 2026.
- [**ICASSP'26**] Shi, J., Du, H., Hong, Y. A., Gao, Y., "EMO-TTA: Improving Test-Time Adaptation of Audio-Language Models for Speech Emotion Recognition." IEEE ICASSP, 2026.
- [**ICASSP'26**] Shi, J., Du, H., Hong, Y. A., Gao, Y., "Plug-and-Play Emotion Graphs for Compositional Prompting in Zero-Shot Speech Emotion Recognition." IEEE ICASSP, 2026.
- [**ICASSP'26**] Mersa, O., Jiang, M., Gao, Y., Li, Q., Zhang, Y., "Hierarchical Convolution Multibranch Transformer for EEG Signals." IEEE ICASSP, 2026.
- [**ICASSP'26**] Jiang, M., Chen, Y., Liu, G., Sahu, A., Gao, Y., Zhang, Y., "Pairs, Not Labels: Predicting Protein-Phenotype Associations via Link Prediction." IEEE ICASSP, 2026.
- [**EMNLP'25**] Du, H., Shi, J., Myerston, J., Lu, S., Zhou, G., Gao, Y., "Role-Guided Annotation and Prototype-Aligned Representation Learning for Historical Literature Sentiment Classification." Findings of EMNLP, 2025.
- [**InterSpeech'25**] Du, H., Lu, S., Zhou, G., Gao, Y., "Emotion-Aware Audio Large Language Models with Dual Cross-Attention and Context-Aware Instruction Tuning." InterSpeech, 2025.
- [**InterSpeech'25**] Shi, J., Zhang, Y., Gao, Y., "Contrastive Learning for Speech Emotion Domain Generalization via Soft Prompt Tuning." InterSpeech, 2025.
- [**CHASE'25**] Clapham, J., Zhou, M., MacDonald, C., Koltermann, K., Gao, Y., Shao, H., "ElectroMeter: The Practical Electrolyte Measurement System." IEEE/ACM CHASE, 2025.
- [**JMIR'25**] Ko, E., Gao, Y., Wang, P., Wijayasingha, L., Wright, K. D., Gordon, K. C., Wang, H., Stankovic, J. A., & Rose, K. M. "Challenges to recruiting dementia caregiving dyads in community-based settings." JMIR Formative Research, 2025.
- [**TMLR'24**] Macdonald, C., Chu, Z., Stankovic, J., Zhou, G., Shao, H., Gao, Y., "MiddleGAN: Generating Inter-domain Images with GANs." Transactions on Machine Learning Research, 2024.
- [**CHASE'24**] Koltermann, K., Clapham, J., Blackwell, G., Jung, W., Burnet, E., Gao, Y., Shao, H., et al., "Gait-Guard: Turn-aware Freezing of Gait Detection for Non-intrusive Intervention Systems." IEEE/ACM CHASE, 2024. (**Best Paper Award**)
- [**SmartComp'23**] Gao, Y., Baucom, B., Gordon, K., Rose, K., Wang, H., Stankovic, J. "E-ADDA: Unsupervised Adversarial Domain Adaptation Enhanced by a New Mahalanobis Distance Loss." IEEE SmartComp, 2023.
- [**HEALTH'21**] Gao, Y., Salekin, A., Gordon, K., Rose, K., Wang, H., Stankovic, J. "Emotion Recognition Robust to Indoor Environmental Distortions Using Out-of-Distribution Detection." ACM Transactions on Computing for Healthcare, 2021.
- [**PERCOM'21**] Gao, Y., et al. "Out-of-the-Box Deployment to Support Research on In-Home Care of Alzheimer's Patients." IEEE Pervasive Computing, 2021.
- [**JAN'21**] Rose, K. M., et al., Gao, Y., "Smarthealth technology study protocol for dementia caregiving relationships." Journal of Advanced Nursing, 2021.
- [**SENSYS'20**] Gao, Y., et al. "A Monitoring and Recommendation System for In-Home Caregivers (Demo)." ACM SenSys, 2020.

Full publication list: <https://scholar.google.com/citations?user=XvbRi5wAAAAJ&hl=en>

THE GALATEA LAB

Mentoring Ph.D. Students

M3. Jiacheng Shi, Ph.D. Student, William & Mary.
First author publications: ICASSP, ACL, Interspeech 2024 - Present

M2. Hongfei Du, Ph.D. Student, William & Mary.
First author publications: ICML, ACL, EMNLP, Interspeech 2024 - Present

Mentoring Undergraduate Students

M1. Lilly Liu, Undergraduate Student, William & Mary.
Initial placement: Carnegie Mellon University (CMU). 2024 - Present

Selected Research Directions

D3. Domain-Generalized Audio Sensing — robust speech emotion recognition under distribution shift

D2. Sentiment Steering in LLMs — controllable generation of emotional tone in language models

D1. Audio Stress Detection for ADRD Caregivers — mobile and scalable systems for caregiver support ([Linda Gaw][1])

HONORS AND AWARDS

Research Awards

H7. Best Paper Award, IEEE/ACM CHASE. 2024

H6. Best Paper Runner-Up Award, IEEE SmartComp. 2023

H5. Best Poster Award, Computer Science Symposium, University of Virginia. 2022

Service Awards

H4. Best Service Award, IEEE/ACM CHASE. 2024

Fellowships and Academic Honors

H3. Ph.D. Fellowship, Computer Science, University of Virginia. 2019

H2. Academic Excellence Fellowship, Computer Science, University of Virginia. 2017

H1. Honors with High Distinction, University of California, San Diego. 2016

REVIEWER SERVICE

R6. Primary Reviewer. NeurIPS. 2026

R5. Primary Reviewer. ICML, AAAI, ICLR. 2025

R4. Primary Reviewer. NeurIPS, AAAI, ICLR, ACM Health. 2024

R3. Primary Reviewer. AAAI, NeurIPS, ICLR. 2023

R2. Primary Reviewer. IEEE Transactions on Affective Computing; AAAI; Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT/UbiComp); NeurIPS. 2022

R1. Primary Reviewer. ACM Transactions on Computing for Healthcare. 2021

PROFESSIONAL MEMBERSHIPS

P2. Association for Computing Machinery (ACM). Member. 2025 - Present

P1. Institute of Electrical and Electronics Engineers (IEEE). Member. 2025

PUBLISHED CREATIVE WRITING

Books

C3. *The Tempest's Call*. Pegasus Publishers, United Kingdom. (To Appear) 2026

C2. *The Celestial Blade*. Pink Hydra Press, South Africa. (To Appear) 2026

Short Fiction

C1. *The Mother of Gaia*. Bewildering Stories, Canada.

2025

FEATURED

F1. Honors student of the Department of Literature at UCSD. [Webpage]

2016

REFERENCES

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Page duBois

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